

Investigating the Impact of Occupancy Behaviour on Predicted Electrical Demand and Energy Flexibility for UK Households

Benjamin Honnywill supervised by Dr Nima Gerami Seresht

Abstract—Building energy use accounts for approximately 30% of global energy consumption. Improving building energy efficiency is therefore critical to help reduce carbon emissions and mitigate climate change. To achieve this, it is key to be able to first accurately forecast energy consumption. Currently, predictive models often only consider the physical and functional characteristics of the building envelope and natural surroundings whilst largely disregarding occupants' energy behaviour (OEB). This simplification can result in significant prediction inaccuracies, reducing their effectiveness. OEB can be challenging to fully implement as it consists of varied and changing factors. This study addresses this by comprehensively quantifying OEB as the difference between simulated and measured building energy use. Fuzzy c-means (FCM) clustering is then applied to categorise OEB into five distinct consumption profiles, which are used to improve energy demand predictions. The resulting OEB-adjusted forecasts were then incorporated into a Mixed-Integer Linear Programming (MILP) and Particle Swarm Optimisation (PSO) framework to optimise the capacity of a communal Battery Energy Storage System (BESS) serving six households. An optimal battery capacity of 99.5kWh was identified, which achieved approximately a 22% reduction in household electricity costs. The system demonstrated strong financial viability, with a positive Net Present Value (NPV) of £32,700 over a 10-year period. This research demonstrates that incorporating comprehensive and categorised OEB into energy prediction models can enhance the design and financial performance of communal energy flexibility systems.

Index Terms—Occupancy Energy Behaviour, Fuzzy Set Theory, Energy Flexibility, Battery Energy Storage System Optimisation

I. INTRODUCTION

GLOBAL energy demand continues to increase each year on average, with building energy use accounting for approximately 30% of total worldwide energy consumption [1]. As the impacts of climate change intensify, improving the energy efficiency of buildings has become increasingly important to reduce greenhouse gas emissions and meet sustainability targets. Accurately predicting energy usage is therefore vital to enable better planning and management.

Various models have been developed to predict a building's energy consumption. However, most focus on the physical and functional characteristics of the building envelope, along with environmental conditions. Physical attributes include factors like the location and thickness of walls, while functional characteristics include properties such as U-values and thermal mass. Environmental conditions include external variables like outdoor air temperature and humidity.

While these characteristics and conditions are important, the occupants' energy behaviour (OEB) also significantly contributes to a building's energy consumption [2], and should therefore be appropriately accounted for in predictive models. OEB encompasses all actions taken by building occupants that influence energy usage. These include direct actions, such as adjusting thermostat settings, as well as indirect actions like clothing choices, which impact the perceived building conditions.

Accurately integrating OEB into predictive models can be challenging due to its variability. People of different ages, cultures and backgrounds often have significantly varied energy usage behaviours. These characteristics can range widely in both timing and magnitude, resulting in large discrepancies compared to standard predicted schedules.

For example, in UK student housing, 85% of residents open a window to cool their room while 34% lower the heating and 23% wear cooler clothes [3]. Such varied responses can contribute towards differences of 300% between modelled and actual building energy consumption [4]. It can therefore be an overly simplistic assumption that occupant energy usage should be modelled identically. Despite this, many predictive models simplify OEB due to the added complexity, which ultimately can reduce their effectiveness.

This paper addresses this challenge by comparing the hourly energy usage between a simulated and actual building to comprehensively quantify OEB. However, due to the uncertain and variable nature of occupant behaviour, it would be unreliable and inaccurate to simply reuse historical OEB values for future predictions.

To account for this uncertainty, this paper applies a fuzzy c-means (FCM) clustering technique to categorise the OEB data into distinct consumption level groups. This approach captures both the variability of future OEB and the subjective uncertainty of ambiguous group boundaries. It does so by allowing each data point to simultaneously belong to multiple groups with varying degrees of membership. The quantified and categorised OEB can then be incorporated into energy predictions, increasing their accuracy and ability to develop energy flexibility strategies.

Energy flexibility is the extent to which a user can adapt their energy consumption or purchasing behaviour. Time-of-use tariffs offer variable hourly electricity rates which can help improve this flexibility. A Battery Energy Storage System (BESS) can further enhance energy flexibility by allowing users to strategically charge a battery when prices are low

and discharge it when prices are higher, a practice known as energy arbitrage.

Integrating OEB into energy predictions can improve their accuracy which allows for better alignment of battery charging and discharging between household energy demand and electricity prices. This can lead to lower energy bills and enhance the overall efficiency of BESS operation. These benefits can be further increased when implemented at a community level through a shared BESS which is appropriately sized for the aggregated demand of multiple households. A communal BESS can also offer economic advantages, as the capital investment cost can be shared among households [5] and a single larger system can be installed instead of several smaller systems, avoiding duplicated installation and peripheral costs.

Despite the potential, there is currently a lack of research that optimises the capacity of a communal BESS while comprehensively considering categorised OEB. This study addresses that gap by applying a FCM clustering algorithm to categorise OEB and improve energy demand predictions. These predictions are then used within a Mixed-Integer Linear Programming (MILP) model, which optimises the BESS charge and discharge cycles based on the load and dynamic pricing. To determine the most cost-effective battery capacity, Particle Swarm Optimisation (PSO) is also applied. For each PSO iteration, the MILP model is run to optimise the BESS scheduling for the candidate battery capacity, enabling PSO to evaluate both operational savings and capital costs and thereby identify the optimal battery size. The overall effectiveness of this approach is analysed using the aggregated electricity demand of six UK households, aiming to improve energy flexibility at the community scale.

The structure of this paper is as follows: Section II presents a literature review covering predictive modelling techniques for building energy consumption, the role of fuzzy set theory in capturing OEB and the application of BESS to improve energy flexibility. Section III details the methodology used to quantify and categorise OEB through fuzzy clustering and incorporates these results into a MILP and PSO framework to optimise communal BESS capacity. Section IV presents and discusses the outcomes of the OEB categorisation and the optimisation results. Finally, Section V summarises the main findings, highlights the study's limitations and suggests directions for future research.

II. LITERATURE REVIEW

This section reviews the existing literature on predicting residential energy consumption and the use of BESS to improve energy flexibility. It first examines how traditional modelling techniques often fail to fully account for OEB and how, despite its potential, FCM clustering has been underused to address behavioural uncertainty. The review then explores how BESS has been used to support energy flexibility, before identifying a key research gap that this paper aims to address: the optimisation of communal BESS capacity remains insufficiently explored whilst comprehensively considering OEB.

A. Predictive Energy Modelling and the Role of Fuzzy Set Theory in Capturing Occupant Behaviour

The energy consumption of residential buildings is influenced by a variety of interrelated factors, including its physical characteristics, environmental conditions and appliance usage. For instance, wall insulation levels and outdoor air temperatures can significantly affect heating and cooling demands and therefore overall energy use. Another key influence is OEB, which includes variables such as the number of occupants, their daily routines and specific actions like thermostat adjustments or leaving lights on [6]. Accurate energy consumption predictions are vital for optimising energy flexibility strategies [7], and should therefore comprehensively account for OEB.

Predictive models for building energy consumption can be categorised into the following types: white-box models, black-box models and grey-box models [8].

White-box models are physical energy models that rely on the fundamental laws of physics and thermodynamics [9]. They predict energy consumption by applying the conservation laws of mass, energy and momentum to the building's envelope parameters and environmental conditions. Empirical energy data is not required, which can be advantageous if collecting or accessing data is impractical. White-box models are considered to be one of the most accurate model types [6] however a limitation is that they struggle to represent OEB, as it is infeasible to explicitly model the full complexity and variability of human actions.

Black-box models are data-driven and use empirical data relating to the building's physical form, weather and appliance usage to predict energy consumption. The available data is typically divided into training and testing sets to develop and validate the model. They allow flexibility in considering OEB however only if there is relevant and high-quality data available. As these models are extremely context specific, they are only directly applicable to the specific building they were developed for [9].

Grey-box models are hybrid models that combine the scientific and data focused approach of white- and black- box models. The conservation equations are simplified to reduce computational complexity and empirical data is then used to calibrate the model. Consequently, the OEB limitations are consistent with the previous models.

As outlined above, existing prediction models routinely oversimplify OEB due to its complexity and variability. In many cases, only isolated aspects of OEB are considered such as occupancy scheduling [10], window opening [11], air conditioning [12] or lighting [13]. However, excessive simplifications can significantly reduce model accuracy, with discrepancies of up to 300% reported between predicted and actual consumption [4], due to the broad range of occupant behaviours [14].

As demonstrated in [6], it is possible to represent OEB by the discrepancy between actual and simulated building energy consumption. However, the variability of occupant behaviour makes it unsuitable to directly reuse measured OEB values for future forecasting. Fuzzy set theory can provide a structured approach to address this uncertainty by categorising OEB into

consumption level groups, which can then be used to inform future predictions.

Fuzzy set theory is a mathematical framework that can help capture subjective uncertainty that arises from ambiguity and vagueness. Unlike random uncertainty, which deals with stochastic variability and is often handled through probability theory, subjective uncertainty is typically based on human perception and linguistic vagueness. In classical set theory, an element either fully belongs or does not belong to a set. Fuzzy set theory extends this concept by allowing elements to partially belong to multiple sets simultaneously, with varying degrees of membership. This flexibility makes fuzzy sets particularly suited to modelling complex systems where category boundaries are not clearly defined. A fuzzy set F can be defined by a fuzzy membership function (FMF) that maps the universe Z 's elements to the interval $[0, 1]$, as seen in Equation 1 [15].

$$F : Z \rightarrow [0, 1] \quad (1)$$

Therefore, each element's degree of membership in a fuzzy set can be represented by the real number $\mu \in [0, 1]$ [6]. This makes fuzzy set theory particularly useful when considering energy behaviour, where the boundaries between categories such as low, medium and high energy use are not sharply defined.

FMFs can be developed using various approaches. A data-driven technique is fuzzy clustering which is a type of unsupervised machine learning that groups data points into multiple clusters based on their similarity. FMF can also develop different shapes. As they can be easy to define and flexible, Gaussian FMFs are often developed from fuzzy clustering algorithms [16]. Among the available techniques, FCM clustering is widely applied in engineering applications and is used in this study to help quantify and categorise OEB [17].

While fuzzy set theory has been applied in recent studies to improve building energy predictions [18] [19], many do not comprehensively consider OEB. Notably, both [6] and [20] use FCM to extensively account for OEB however they do not use five or more clusters to get a broad range of behavioural profiles from comparing actual and simulated building energy consumption. Furthermore, neither study investigates OEB at an hourly resolution, which is key for improving the predictions used in BESS applications. This study aims to address these limitations by integrating high-resolution, clustered OEB into the capacity optimisation of a communal BESS.

B. Optimising Energy Flexibility with Battery Storage Systems

Accurate prediction of electrical load demand has a wide range of applications [21], including the development of energy flexibility strategies in UK residential buildings [22].

The use of BESS to improve energy flexibility has been widely studied [23]. Previous research has applied MILP to optimise BESS charge and discharge cycles [24] [25] and PSO to determine optimal BESS capacity [26]. Additionally, the potential of communal BESS has been explored [5].

PSO is an algorithm that mimics the social behaviour of animals, such as a flock of birds, to explore complex

optimisation spaces [27]. It works by evolving a population of potential solutions (particles) through iterative updates based on individual and collective experience. Its ability to efficiently search optimisation spaces makes it well-suited for solving problems such as battery sizing.

However, there is limited research that combines these optimisation techniques (MILP for scheduling and PSO for sizing) while also comprehensively considering OEB in the context of communal BESS. This study addresses this gap by applying FCM clustering to characterise OEB, which is then integrated into the optimisation of a shared BESS serving six UK households. In doing so, it investigates the impact of OEB on community-scale energy flexibility.

III. METHODOLOGY

This section details the methods used to quantify, categorise and evaluate the impact of OEB on energy flexibility in a communal residential BESS. It is divided into three parts: (A) simulating energy demand and comparing it with measured data to quantify OEB, (B) categorising OEB using FCM clustering and (C) applying an optimisation framework that combines MILP and PSO to determine the optimal capacity of a shared BESS when OEB is considered.

A. Quantifying Occupancy Energy Behaviour from Simulated and Measured Data

This study quantifies OEB by comparing the energy consumption of a simulated building with measured data from a similar existing building.

Energy simulations were performed using EnergyPlus (EP), a widely used white-box modelling engine developed by the U.S. Department of Energy (DOE). DOE reference building models were used for the simulations, as they were developed to be a foundation for researching energy efficiency in buildings [28]. The measured energy consumption data were sourced from the Building Data Genome Project 2's open dataset that contains hourly electricity meter readings for non-residential buildings across the UK during 2017 [29].

While this paper focuses on residential buildings for the communal BESS optimisation, measured residential data were not readily available. As a result, OEB was estimated using medium-sized office buildings ($4825m^2 \pm 3.8\%$) [30] from the Genome dataset, and subsequently applied to a DOE reference single-family detached house model [31]. This substitution is justified by the assumption that an individual's energy-consuming behaviours are broadly consistent between work and home environments, making office-derived OEB a reasonable estimate for residential scenarios.

To ensure compatibility, all simulations and measured data corresponded to ASHRAE Climate Zone 4A, which represents London's environmental conditions, and energy consumption values were normalised by floor area. Due to its white-box modelling approach, the base simulation was assumed to provide an accurate representation of building energy use, apart from OEB. Therefore, as both the simulated and measured buildings were of the same type (medium-sized offices), the difference in energy use can be attributed to OEB. The

resulting hourly per-square-metre OEB relationship for each of the six measured buildings in comparison to the simulated baseline can be expressed by Equation 2 [6].

$$OEB \left(\frac{kWh}{m^2} \right) \propto \left(\frac{M_p}{A_p} \right)_t - \left(\frac{S_q}{A_q} \right)_t \quad (2)$$

where M is the measured consumption, S is the simulated consumption, A_p and A_q are the floor areas of the respective buildings and t denotes the time period. The subscripts p and q refer to individual buildings.

The simulated model accounts for the impact of occupancy by each zone's average number of people alongside a predicted schedule. In contrast, the Genome data is measured energy use and therefore entirely includes the effects of OEB. While there is no explicit data about individual behavioural actions, the discrepancy between the measured and simulated consumptions can still quantify OEB.

OEB was evaluated over a full year on an hourly basis to capture its variability across different seasons and throughout the day. For example, seasonal changes such as colder winters or longer summer days can influence heating, cooling and lighting behaviours, further affecting overall energy use. Similarly, a household might exhibit high energy use behaviours in the morning but lower usage in the evening. This variation, observed across both generally higher and lower consumption buildings, is illustrated in Figure 1 and is important to capture for improving the energy predictions used in BESS optimisation.

B. Categorising Occupancy Energy Behaviour Using Fuzzy C-Means Clustering

A key goal of incorporating OEB into energy prediction models is to increase their accuracy. Since it is impossible to obtain measured data for future energy consumption, Equation 2 can only be used to quantify historical OEB. However, due to the variability of human behaviour, it is unrealistic to assume that the OEB for a specific hour (e.g., midday on 28

February 2025) will be identical in a future year (e.g., midday on 28 February 2026).

Clustering the OEB data points into consumption level categories allows each hour to be represented by a behavioural group (e.g., high consumer) that remains constant in future predictions. It is assumed that, while OEB may vary from year to year, it will remain within the same general consumption group. Although each cluster contains a range of OEB values, the data points within each group are broadly similar. Each cluster can therefore be assigned a representative value, based on the spread of OEB values within the group, to replace the original OEB data points in energy predictions. This approach systematically incorporates reasonable future variability of OEB into the predictive process.

Specifically, fuzzy clustering was applied because there is subjective uncertainty in the boundaries between different consumption groups. Traditional crisp clustering would assign data points strictly to a single group — for example, an OEB value of $0.099 kWh/m^2$ could fall into the "high" category, while a value of $0.100 kWh/m^2$ could be classified as "very high", despite their near-identical magnitudes. In contrast, fuzzy clustering recognises that such values are very similar and allows partial membership across multiple groups, better capturing the gradual transitions observed in real occupant behaviour.

The hourly OEB data points for all buildings, calculated using Equation 2, were categorised into the energy consumption profiles using MATLAB's FCM clustering `genfis` function.

Equation 3 defines the objective function Q that FCM minimises to group OEB data points into clusters. It does so by reducing the weighted distance between data points and cluster centres, while allowing partial memberships to capture the uncertainty in OEB profiles [15]:

$$Q = \sum_{c=1}^C \sum_{k=1}^N \mu_{ck}^m \|x_k - v_c\|^2 \quad (3)$$

where:

- C is the number of clusters

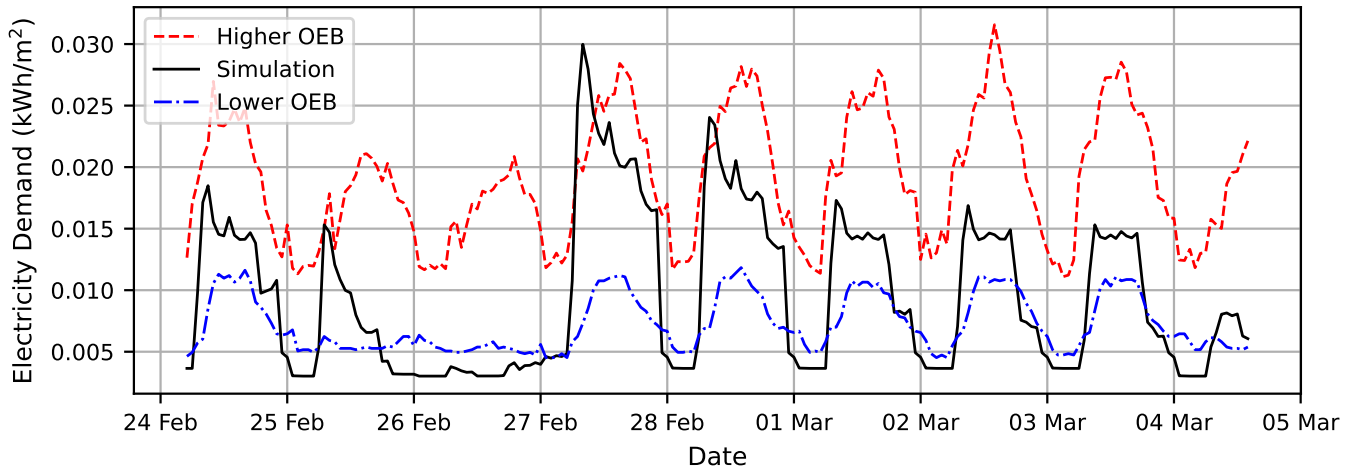


Fig. 1. A period of the annual EP simulated electricity demand and two example higher and lower consumption offices from the Genome dataset.

- N is the number of data points
- μ_{ck} is the membership degree of the k^{th} data point x_k to cluster c
- v_c is the centre of cluster c
- m is the fuzzification coefficient, which impacts how strongly data points are associated with multiple clusters
- $\|x_k - v_c\|$ is the distance between x_k and v_c

The FCM clustering algorithm developed Gaussian FMFs, as shown in Figure 2. The figure illustrates how the degree of membership for each OEB data point varies depending on the cluster. Due to the subjective uncertainty from the ambiguity of boundaries between energy consumption profiles, a degree of overlap between the FMFs is appropriate and can be observed.

The fuzzification coefficient m influences the extent of overlap between clusters [15]. As $m \rightarrow 1$, the FMFs tend towards each data point belonging exclusively to a single cluster. This is consistent with classical set theory and is typical of techniques such as K-means clustering. On the other hand, values over 2 can result in spiky FMFs. Therefore, in this study, m was set to 2 to effectively capture the subjective uncertainty between group boundaries while maintaining sufficient separation between clusters. This choice allows for clear linguistic labels to be assigned, which is beneficial for OEB categorisation and therefore energy predictions.

The number of clusters was set to five based on several key considerations. A primary factor was the principle of justifiable granularity, which requires a concise yet meaningful representation of the data. This principle emphasises that each cluster should be well-supported by the data while remaining distinct from the others. Typically, it is suggested that a suitable number of clusters is within the range of five to nine [15]. Furthermore, selecting an odd number of clusters allows for a central cluster, which can serve as a neutral reference point that helps to interpret the data. The five clusters were qualitatively interpreted using linguistic labels: very low, low, medium, high and very high energy consumers. These labels enable an intuitive understanding of the subjective uncertainty

from categorising OEB, while avoiding overly fine distinctions such as "very very low consumer".

The cluster representative values were used to account for future variability in OEB within the predicted demand. They were determined based on a membership value confidence level, which defined the minimum degree of belonging required for a data point to be considered reasonably part of a behavioural cluster. This level established a corresponding confidence interval for each cluster. The maximum OEB value within each cluster, determined after applying the confidence level, was selected as the upper bound of the interval and could then be used as the representative value for data points in that cluster for the predicted demand. This approach was adopted to improve the robustness of energy forecasts by incorporating a realistic upper limit of OEB, thereby enabling the BESS to be more effectively scaled and operated to accommodate a range of demand scenarios.

If a data point had a membership value exceeding the confidence level for multiple clusters, a fuzzy set operation was used to determine its representative value. Specifically, the maximum t-conorm (standard union) was applied, assigning the highest consumption cluster representative value to the data point to ensure the full acceptable variability of OEB was captured. If a data point did not meet the confidence level for any cluster, it was treated as an anomaly and assigned a final predicted OEB value of zero. This situation occurred in 0.2% of the data points under the applied confidence level of 0.3.

The confidence level of 0.3 was selected based on the use case of a communal BESS and to capture a reasonable amount of variance in future OEB. Since the predicted demand load for optimising the BESS was the hourly aggregate of six households, the possibility of the combined load simultaneously exceeding the upper bounds of their respective cluster OEB predictions was minimal. For example, while some households might occasionally have OEB above their cluster's upper limit, it is likely that others would remain below theirs, effectively balancing the overall demand. Additionally, inspection of the FMF distributions in Figure 2 showed that a confidence

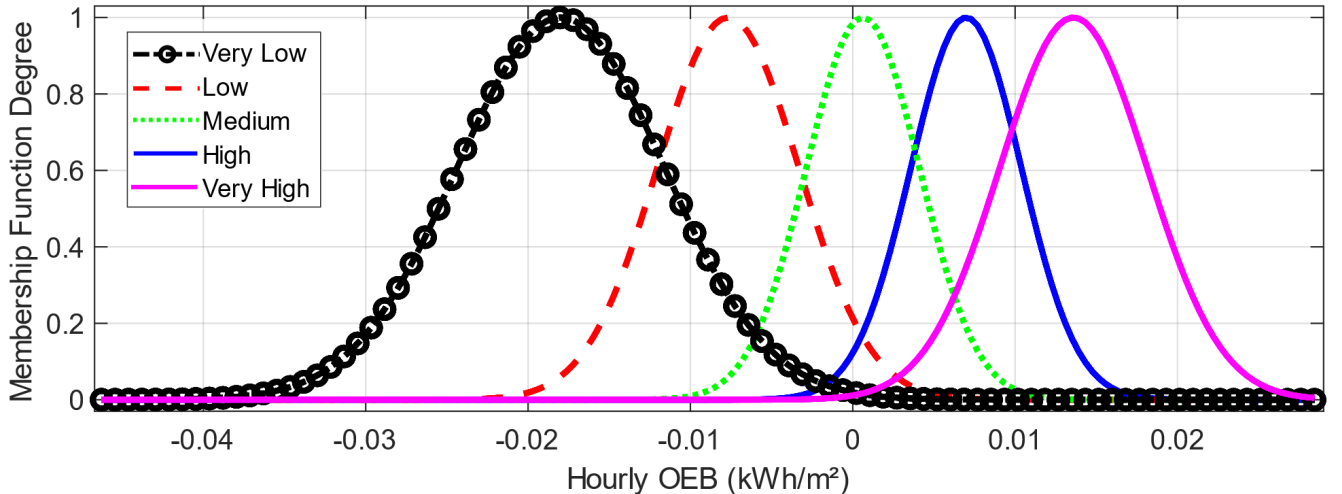


Fig. 2. Gaussian Fuzzy Membership Functions (FMFs) for Categorised Occupants' Energy Behaviour (OEB) Clusters.

level of 0.3 ensured that data points were not simultaneously categorised into three distinct clusters (e.g., low, medium, and high consumers), which would have undermined the clarity of the groupings. Although a lower confidence level could have been used, excessive accounting for OEB uncertainty would likely have reduced the accuracy of predictions. Overall, an appropriate balance was important to produce accurate predictions for optimising the BESS capacity and improving energy flexibility.

C. Optimising Communal BESS Capacity Using MILP and PSO

Building on the categorised OEB predictions developed in the previous section, this part of the methodology focuses on optimising the capacity of a communal residential BESS to improve energy flexibility.

The predicted hourly energy load for the BESS was based on the aggregation of six residential buildings. Each building used the same base EP simulation model of a DOE prototype single-family detached house [31] and then the categorised OEB predictions were added. A BESS sized for six households was selected, as this was considered a suitable scale for evaluating community-level energy flexibility.

Octopus Energy's Agile Import Tariff Hourly Average for the East Midlands, UK, was used for the variable electricity price [32]. This tariff was selected due to its time-of-use pricing structure, which enables effective energy arbitrage.

Apart from the capacity, the BESS physical specifications were fixed and primarily based on a Tesla Powerwall 3 [35] as this was an appropriate and existing BESS option available on the market.

The battery capacity of the BESS was optimised by combining PSO with a MILP model. In each PSO iteration, the MILP model was run to determine the optimal hourly charging and discharging schedule for the candidate battery capacity,

allowing PSO to evaluate operational performance alongside capital costs and identify the most cost-effective system size.

The MILP model aimed to minimise total energy costs by performing energy arbitrage. This optimisation was based on balancing the predicted hourly demand and electricity prices. The MILP implementation was adapted from publicly available code authored by Yeap Yew Ming [36], with additional reference to the modelling framework outlined by Eric M. [37].

The power system was modelled as the aggregated predicted load of the residential buildings, a communal BESS and the national grid. The parameters listed in Table I were used to define the MILP constraints shown in Equations 4–10.

It was essential to maintain power balance at each time period between the grid, predicted load and BESS, captured by:

$$P_{grid}^t = P_{load}^t - P_{bat}^t \quad (4)$$

where P_{grid}^t is positive when importing and negative when exporting.

To respect UK regulations on grid export limits [34], the following constraint was imposed, where the limit was the aggregate for the six houses:

$$P_{bat}^t - P_{load}^t \leq P_{max,export} \quad (5)$$

No import limit was applied, as the expected loads were well within practical grid import capacities.

The BESS charging and discharging operations were split:

$$P_{bat}^t = P_{bat,charge}^t + P_{bat,discharge}^t \quad (6)$$

with charging considered negative and discharging positive.

To prevent simultaneous charging and discharging, a binary variable σ^t was introduced to represent the battery's charging status. When $\sigma^t = 1$, the battery cannot discharge, and

TABLE I
BATTERY ENERGY STORAGE SYSTEM VARIABLES USED IN THE OPTIMISATION

Symbol / Variable	Description / Type	Value	Units	Reference
t	Time period	$\in \{1, 2, \dots, 8760\}$	hours	–
P_{grid}^t	Power imported/exported from/to grid	Continuous, all real numbers	kW	–
P_{load}^t	Power demand of aggregate load	Continuous, positive real numbers	kW	–
P_{bat}^t	Power of the BESS	Continuous, all real numbers	kW	–
$P_{bat,charge}^t$	BESS charging power	Continuous, negative real numbers	kW	–
$P_{bat,discharge}^t$	BESS discharging power	Continuous, positive real numbers	kW	–
σ^t	Charging status	Binary (0 or 1)	–	–
SOC^t	State of Charge of BESS	Continuous, 0–1	%	–
SOC_{min}, SOC_{max}	Minimum/maximum State of Charge	1, 99	%	–
Δt	Time interval	1	hours	–
C_{market}^t	Cost of electricity	Continuous, all real numbers	£/kWh	[32]
$C_{var,bat}$	Cost of battery	90	£/kWh	[33]
$P_{max,export}$	Maximum power exported to grid	22.08	kW	[34]
$P_{bat,charge,max}$	Maximum BESS charging power	(–)23	kW	[35]
$P_{bat,discharge,max}$	Maximum BESS discharging power	23	kW	[35]
η	Charging/discharging efficiency	89	%	[35]
Y	BESS lifetime	10	years	[35]
E	BESS energy capacity	1 to 162	kWh	[35]

when $\sigma^t = 0$, it cannot charge. This behaviour is enforced through the following constraints, which also ensure that the BESS's maximum rated charging and discharging powers are not exceeded:

$$P_{bat,charge}^t \geq \sigma^t P_{bat,charge,max} \quad (7)$$

$$P_{bat,discharge}^t \leq (1 - \sigma^t) P_{bat,discharge,max} \quad (8)$$

The BESS State of Charge (SOC) was updated at each time period by:

$$SOC^t = SOC^{t-1} - \frac{\Delta t}{E} \left(P_{bat,charge}^t \eta + \frac{P_{bat,discharge}^t}{\eta} \right) \quad (9)$$

subject to the SOC BESS bounds:

$$SOC_{min} \leq SOC^t \leq SOC_{max} \quad (10)$$

where SOC_{min} and SOC_{max} were set to 1% and 99% respectively as a simplifying assumption.

The overall minimisation objective of the combined MILP-PSO model is given by Equation 11:

$$\text{minimise } C_{total} = C_{var,bat}E + Y \sum_t P_{grid}^t C_{market}^t \Delta t \quad (11)$$

The main objective of the MILP part of the model was to minimise the electricity cost for the candidate battery capacity in each PSO iteration. Savings from the optimised charging and discharging schedule were calculated over a one-year period and assumed to repeat annually across the BESS lifetime, as represented by the second, summation term in Equation 11.

The MILP model was implemented using Python's open-source `pywraplp` module from `ortools.linear_solver`, with the CBC backend solver.

To optimise the battery capacity, PSO was applied to minimise the total cost C_{total} by considering both the variable battery cost and lifetime energy savings, as seen in the completed form of Equation 11. This was achieved using Python's `pyswarms` library and by using the output from the MILP part of the model.

PSO, as introduced in Section II, first initialises a swarm of particles with random positions and velocities, each representing a potential solution [27]. Each particle is then iteratively evaluated against the objective function (Equation 11) and its best position so far p_{best_i} is stored. When a particle achieves a higher fitness score than any of its previous positions, its current location is updated accordingly. The global best position g_{best_i} represents the best solution found by the entire swarm.

Particles adjust their movement based on both their individual and collective experience, enabling the search for the optimal battery capacity. Their velocities and positions are updated at each iteration according to:

$$v_i(n+1) = \omega v_i(n) + c_1 r_1 [p_{best_i} - x_i(n)] + c_2 r_2 [g_{best_i} - x_i(n)] \quad (12)$$

$$x_i(n+1) = x_i(n) + v_i(n+1) \quad (13)$$

where:

- ω is the inertia weight, determining how much the previous velocity influences the present movement
- c_1 and c_2 are the cognitive and social learning factors
- r_1 and r_2 are random numbers between 0 and 1
- p_{best_i} is the particle's own best-known position
- g_{best_i} is the best-known position across the entire swarm
- $x_i(n)$ and $v_i(n)$ are the position and velocity of particle i at iteration n

The inertia weight ω was set to 0.9, while the cognitive and social learning factors c_1 and c_2 were 0.5 and 0.3 respectively.

The PSO process was repeated until the termination criteria were met. In this study, early stopping was applied after 10 iterations if the solution did not improve by more than £1. This configuration was selected because the PSO optimisation space was one-dimensional and relatively simple, with testing showing the presence of only one global (or local) minimum. For similar reasons, a modest swarm size of 30 particles was chosen, which was sufficient to achieve quick yet accurate convergence. To validate this choice, a comparison was made against a run using 100 particles, which produced a final solution within £1 of the 30-particle result. This confirmed that a swarm size of 30 was sufficient to reliably reach the optimal solution without unnecessary computational overhead.

Once the optimal capacity was determined, Net Present Value (NPV) analysis was used to evaluate whether a communal BESS investment would be financially viable. It can provide a measure of the opportunity cost of investing by discounting future savings to their present value. The NPV was calculated according to:

$$NPV = \sum_{y=0}^Y \frac{C_y}{(1+d)^y} \quad (14)$$

where C_y is the net cash flow at year y , Y is the project lifetime in years and d is the discount rate [38].

A discount rate of 4.5% was applied, based on a recent Bank of England base rate [39]. In addition to the optimised battery cell cost, a fixed installation cost of £2500 was included to cover installation and additional hardware components [40]. This reflects the cost advantage of using a communal BESS, as a single larger system avoids the duplicated costs associated with installing multiple smaller systems. For simplicity, maintenance costs, taxes and profit margins were excluded.

IV. RESULTS AND DISCUSSIONS

This section presents and discusses the outcomes from the OEB quantification and categorisation process, and evaluates the impact of incorporating OEB into predictions used for optimising a communal BESS with a MILP-PSO framework.

A. Impact of Occupancy Energy Behaviour on Load Predictions

The hourly OEB, calculated using Equation 2, over a full year across six buildings resulted in 52,560 data points to be clustered by the FCM algorithm for use in energy predictions.

The data points were reasonably distributed across the five fuzzy clusters. The majority fell into the medium (26.9%), high (25.8%) and very high (22.9%) consumption categories, indicating that intensive energy behaviours were more common. In contrast, low (15.9%) and very low (8.4%) behaviours were less frequent which reflects fewer instances of conservative energy use. The spread supports the selection of five clusters, as each was meaningfully populated while capturing a wide range of occupant behaviour.

The baseline simulation for the single-family detached house produced an annual electricity consumption of $75kWh/m^2$. The house was heated and cooled by a heat pump and research indicates that the median electricity consumption for an all-electric home in the UK is $87kWh/m^2$ [41]. This considers a variety of building types, including single-family houses, apartment blocks and collective housing. In theory, having a higher external wall area/living space ratio, like detached single-family houses, will be less energy efficient [42]. As a result, their electricity consumption is likely to exceed the upper interquartile range of $129kWh/m^2$. Since the simulated baseline value is significantly lower, this highlights the importance of incorporating OEB to produce more representative energy predictions.

When the clustered OEB values were incorporated into the baseline prediction, the resulting predicted annual electricity use increased to $170kWh/m^2$. While this value exceeds the upper interquartile range reported in the literature, it remains below the upper range of $210kWh/m^2$ (which excludes outliers). The OEB contributed to an average annual increase of 125%, which is consistent with findings in previous studies [4]. As the added OEB values were derived from the upper bounds of the clusters to capture behavioural variability, the predicted energy use represents an upper reasonable estimate of potential demand. This is important for appropriately scaling and operating the BESS however actual household demand could be lower in practice.

It is also important to note that because the OEB adjustments varied hourly, improvements in the hourly resolution of predictions were critical for enhancing the effectiveness of BESS energy arbitrage.

B. Optimisation Results and Financial Evaluation of the Commercial BESS

The MILP's generated scheduling performed effective energy arbitrage, resulting in the battery charging during lower price periods and discharging during higher price periods. This output was then reliably used within the PSO algorithm to evaluate candidate BESS capacities and determine the optimal value.

A battery capacity of $99.5kWh$ was found to be optimal, as shown in Figure 3. When divided across the six households, the average capacity per house aligns well with existing

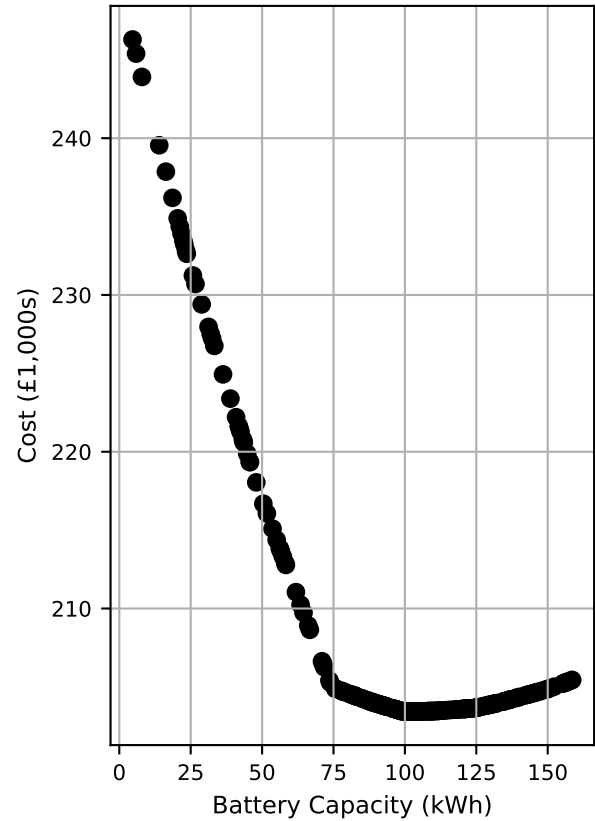


Fig. 3. Visualisation of the PSO optimisation space for battery capacity sizing. Each point represents the evaluated total cost for a candidate battery capacity.

commercial market options [35]. This capacity effectively balanced the trade-off between investment cost and operational savings over a 10-year period. At small battery capacities, total costs are high due to limited ability to perform energy arbitrage and capitalise on low electricity prices. As capacity increases, the total cost initially falls sharply as operational savings improve. At larger capacities, the battery becomes oversized relative to the aggregated household demand and maximum charge/discharge rates, so the additional investment cost increases the total cost.

However, it is important to note that the DOE baseline household model used for the energy demand simulations had a floor area of $220m^2$, which is larger than the average UK detached house of $149m^2$ [43]. Although energy values were standardised by area when quantifying OEB, the final aggregate load for the BESS optimisation reflected the larger model size and consequently higher overall demand. This could potentially have skewed the optimal BESS capacity and the associated savings. This could be addressed by developing a custom residential model however this was beyond the scope of the project.

The optimised BESS resulted in an annual electricity bill reduction of approximately 22%, approaching the lower bound of savings reported in the literature, where reductions of around 28% were observed under certain scenarios [44]. The slightly lower savings achieved in this study could be attributed

to the specific combination of battery capacity, load magnitude and the OEB data.

An annual energy cost saving of approximately £920 per household was achieved using the BESS arbitrage. When viewed as a 22% saving relative to household electricity consumption, this figure appears higher than typical expectations based on UK average energy bills [45]. However, the result is reasonable because it reflects a maximum potential saving, based on an upper estimate of demand used to ensure the BESS was robustly sized and operated. Furthermore, the modelled house had a relatively large floor area, increasing total energy consumption. If the demanded energy was lower, the savings would also likely be reduced.

The NPV of the investment was calculated as £32,700. The total initial cost was £11,455, comprising of £8955 for the optimised 99.5kWh battery capacity (at a capital cost of £90/kWh [33]) and an additional £2500 for installation [40].

The resulting net annual cash flow was approximately £5520, derived from the difference between the annual electricity costs with and without the BESS (£19,442 and £24,962 respectively). As noted, the magnitude of these values reflects the high consumption levels driven by both the upper bound OEB adjustments and the larger house size.

The positive NPV indicates that the BESS would be financially viable under the assumptions made. This highlights the economic potential of leveraging communal energy flexibility, particularly when occupant behaviour is explicitly incorporated into demand prediction models.

V. CONCLUSION, LIMITATIONS AND FUTURE WORK

This study addresses the challenge of incorporating OEB into energy demand predictions to improve community-level energy flexibility.

A review of the literature highlighted that existing building energy prediction models often fail to comprehensively capture the impact of OEB, limiting their accuracy. As OEB is complex and influenced by multiple, varying factors, modelling it accurately can be challenging. However, this paper has comprehensively quantified the effect of OEB as the difference between simulated and measured building energy consumption.

The FCM clustering algorithm was then used to categorise hourly OEB values into consumption level groups which were subsequently used to improve the energy demand predictions. Five clusters were found to effectively capture the range and uncertainty of OEB. These clusters were well-supported by the data and sufficiently distinct from one another. They were labelled as very low, low, medium, high and very high energy consumers. Incorporating these adjusted predictions increased the annual simulated demand by 125%, producing a more realistic load profile that better represents potential future conditions.

The enhanced demand predictions were then applied within a MILP-PSO framework to optimise the sizing of a communal BESS. A battery capacity of 99.5kWh was identified as optimal for the six aggregated households, balancing operational savings with investment cost. This configuration achieved

an approximate 22% reduction in annual electricity costs, corresponding to potential savings of around £920 per household per year. Furthermore, the system's financial viability was confirmed by a positive NPV of £32,700, highlighting the economic benefits of improving energy flexibility with a communal BESS while accounting for OEB.

Overall, these findings demonstrate the importance of accurately modelling occupant behaviour in energy predictions to improve the sizing and financial performance of communal residential energy systems.

One limitation of this study is that the OEB was derived from medium-sized offices, and then applied to residential homes. Offices were selected because sufficient high-quality measured data were available, and because occupant-driven behaviours, such as adjusting windows and thermostats, were still reasonably represented. However, the context of residential energy use is different. Behaviours specific to homes, such as cooking patterns, appliance usage and family routines, may not be fully captured, potentially affecting the accuracy of the OEB adjustment when transferred to a residential setting.

Additionally, the modelling of the BESS could be improved. A simplified pricing structure was used, assuming that the hourly electricity prices from the modelled year would remain constant over the 10-year project lifespan. In reality, average prices are likely to vary, and incorporating an export-specific tariff, which is typically lower than import tariffs, would likely have reduced the predicted savings but provided a more realistic economic assessment. Battery degradation over time was also not considered, meaning that battery performance was likely overestimated across the project lifetime. In addition, maintenance requirements were not included, which could affect the financial viability and reliability of the BESS.

Future work could focus on refining the representative cluster values used for incorporating OEB into energy predictions. Implementing lower bound confidence intervals, in addition to upper bounds, would allow both the minimum and maximum reasonable OEB values to be considered. This could produce a more balanced and realistic predicted demand. Additionally, alternative clustering methods, such as subtractive clustering or Gustafson-Kessel clustering, could be explored to assess whether they better capture OEB variance compared to FCM.

Another key area for future development would be expanding the functionality of the BESS model to incorporate additional technologies. Adding the ability to model personal solar generation and battery electric vehicles (BEVs) would enhance the versatility of the system. This improvement would ensure the model remains relevant as the energy sector increasingly transitions towards decentralised generation and electrified transport.

ACKNOWLEDGMENTS

The author would like to thank Dr Nima Gerami-Seresht for his supervision and support throughout this project. It is also acknowledged that ChatGPT was used for improving the readability of text and efficiency of code.

REFERENCES

- [1] International Energy Agency, "Tracking Clean Energy Progress 2023," <https://www.iea.org/energy-system/buildings>, 2023, Accessed January 2025.
- [2] M. Jia, R. Srinivasan, R. J. Ries, G. Bharathy, and N. Weyer, "Investigating the impact of actual and modeled occupant behavior information input to building performance simulation," *Buildings*, vol. 11, no. 1, 2021.
- [3] UTOPI, "The resident report," 2024, accessed: 2025-04-10.
- [4] E. Delzendeh, S. Wu, A. Lee, and Y. Zhou, "The impact of occupants' behaviours on building energy analysis: A research review," *Renewable and Sustainable Energy Reviews*, vol. 80, pp. 1061–1071, 2017.
- [5] R. Asri, H. Aki, and D. Kodaira, "Study of shared energy storage considering battery degradation on remote island," 2024, Conference paper, p. 311 – 315, cited by: 1.
- [6] E. Mahamedi, M. Wonders, W. Woo, and N. Gerami Seresht, "Quantifying occupants' energy behaviour: A novel fuzzy approach," Aug. 2022, pp. 1–8, copyright (c) 2022 University of New Brunswick.; Transforming Construction with Reality Capture Technologies : The Digital Reality of Tomorrow, TCRC2022 ; Conference date: 23-08-2022 Through 26-08-2022.
- [7] Y. Li, Z. O'Neill, L. Zhang, J. Chen, P. Im, and J. DeGraw, "Grey-box modeling and application for building energy simulations - a critical review," *Renewable and Sustainable Energy Reviews*, vol. 146, p. 111174, 2021.
- [8] A. Kathirgamanathan, M. De Rosa, E. Mangina, and D. P. Finn, "Data-driven predictive control for unlocking building energy flexibility: A review," *Renewable and Sustainable Energy Reviews*, vol. 135, p. 110120, 2021.
- [9] D. Coakley, P. Raftery, and M. Keane, "A review of methods to match building energy simulation models to measured data," *Renewable and Sustainable Energy Reviews*, vol. 37, p. 123–141, 09 2014.
- [10] S. Jami, N. Forouzandeh, Z. S. Zomorodian, M. Tahsildoost, and M. Khoshbakht, "The effect of occupant behaviors on energy retrofit: A case study of student dormitories in tehran," *Journal of Cleaner Production*, vol. 278, p. 123556, 2021.
- [11] Y. Liu, W. T. Chong, Y. Cao, H. Liu, H. Yu, T. Cui, L. Chang, and S. Pan, "Characteristics analysis and modeling of occupants' window operation behavior in hot summer and cold winter region, china," *Building and Environment*, vol. 216, p. 108998, 2022.
- [12] M. Esrafilian-Najafabadi and F. Haghighat, "Impact of occupancy prediction models on building hvac control system performance: Application of machine learning techniques," *Energy and Buildings*, vol. 257, p. 111808, 2022.
- [13] M. Ángel Campano, I. Acosta, S. Domínguez, and R. López-Lovillo, "Dynamic analysis of office lighting smart controls management based on user requirements," *Automation in Construction*, vol. 133, p. 104021, 2022.
- [14] S. Chen, G. Zhang, X. Xia, Y. Chen, S. Setunge, and L. Shi, "The impacts of occupant behavior on building energy consumption: A review," *Sustainable Energy Technologies and Assessments*, vol. 45, p. 101212, 2021.
- [15] W. Pedrycz, *An Introduction to Computing with Fuzzy Sets*. Springer, 2020.
- [16] W. Pedrycz and F. Gomide, *Fuzzy Systems Engineering: Toward Human-Centric Computing*, 01 2007.
- [17] J. C. Bezdek, R. Ehrlich, and W. Full, "Fcm: The fuzzy c-means clustering algorithm," *Computers Geosciences*, vol. 10, no. 2, pp. 191–203, 1984.
- [18] F. Al-Shanableh and A. Evcil, "Prediction of energy consumption of residential buildings in northern cyprus using fuzzy interference system," *Energy and Buildings*, vol. 256, p. 111555, 2022.
- [19] J. Pérez-Gelves, P. Castro-Verdezoto, N. Alvarado-Cantos, and T. César, "Applying fuzzy logic and neural networks to forecasting in efficiency programs," *E3S Web of Conferences*, vol. 532, 06 2024.
- [20] H. Wen, X. Liu, M. Yang, B. Lei, C. Xu, and Z. Chen, "A novel approach for identifying customer groups for personalized demand-side management services using household socio-demographic data," *Energy*, vol. 286, p. 129593, 2024.
- [21] M. Choubey, R. K. Chaurasiya, and J. S. Yadav, "Predicting electrical load demands using neural prophet-based forecasting model," *SN Comput. Sci.*, vol. 6, no. 1, Dec. 2024.
- [22] J. Torriti, "Household electricity demand, the intrinsic flexibility index and uk wholesale electricity market prices," *Environmental Economics and Policy Studies*, vol. 24, 12 2020.
- [23] N. G. Chatzigeorgiou, S. Theocharides, G. Makrides, and G. E. Georghiou, "A review on battery energy storage systems: Applications, developments, and research trends of hybrid installations in the end-user sector," *Journal of Energy Storage*, vol. 86, p. 111192, 2024.
- [24] F. Guo, L. Gomes, L. Ma, Z. Tian, Z. Vale, and S. Pang, "Optimizing battery storage for sustainable energy communities: A multi-scenario analysis," *Sustainable Cities and Society*, vol. 118, p. 106030, 2025.
- [25] A. Cosic, M. Stadler, M. Mansoor, and M. Zellinger, "Mixed-integer linear programming based optimization strategies for renewable energy communities," *Energy*, vol. 237, p. 121559, 2021.
- [26] J. Cai, K. Wang, A. D. Cheok, and Y. Yan, "Optimal configuration for power grid battery energy storage systems based on payload fluctuation guided multi-objective pso," *Journal of Energy Storage*, vol. 105, p. 114515, 2025.
- [27] O. Ibrahim, M. J. Abdul Aziz, R. Ayop, A. T. Dahiru, W. Y. Low, M. H. Sulaiman, and T. I. Amosa, "Fuzzy logic-based particle swarm optimization for integrated energy management system considering battery storage degradation," *Results in Engineering*, vol. 24, p. 102816, 2024.
- [28] K. Field, M. Deru, and D. Studer, "Using doe commercial reference buildings for simulation studies: Preprint," *Proceedings of the 4th National Conference of IBPSA-USA*, 01 2010.
- [29] C. Miller, A. Kathirgamanathan, and B. Picchetti, "The building data genome project 2, energy meter data from the ashrae great energy predictor iii competition," *Scientific Data*, vol. 7, no. 368, 2020.
- [30] U.S. Department of Energy, "Office model: ASHRAE901_OfficeMedium_STD2016_NewYork.idf," <https://www.energycodes.gov/prototype-building-models>, accessed: 24 April 2025.
- [31] —, "House model: US_SF_CZ4A_hp_slab_IECC_2024.idf," <https://www.energycodes.gov/prototype-building-models>, accessed: 24 April 2025.
- [32] AgileBuddy, "Octopus energy agile import tariff hourly average for the east midlands, uk," <https://agilebuddy.uk/historic/download>, accessed: 2025-02-27.
- [33] BloombergNEF, "Lithium-ion battery pack prices see largest drop since 2017, falling to \$115 per kilowatt-hour," December 2023, accessed: 2025-02-27. [Online]. Available: <https://about.bnef.com/blog/lithium-ion-battery-pack-prices-see-largest-drop-since-2017-falling-to-115-per-kilowatt-hour-bloombergnef/>
- [34] SP Energy Networks, "Export limitation and control systems," accessed: 2025-02-27. [Online]. Available: https://www.spenergynetworks.co.uk/pages/export_limitation.aspx
- [35] Tesla, Inc., "Powerwall 3 datasheet," <https://www.tesla.com/support/energy/powerwall/documents/documents>, accessed: 2025-02-27.
- [36] Y. Y. Ming, "Batterydispatcher2.0 github repository," <https://github.com/yeap0022/BatteryDispatcherVer2.0>, 2024, available under MIT License.
- [37] E. M., "Basic build optimization model to schedule battery's operation in power grid systems," <https://medium.com/@yeap0022/basic-build-optimization-model-to-schedule-batterys-operation-in-power-grid-systems-51a8c04b3a0e>, 2024.
- [38] W. Short, D. Packey, and T. Holt, "A manual for the economic evaluation of energy efficiency and renewable energy technologies," *NASA STI/Recon Technical Report N*, vol. 95, 03 1995.
- [39] Bank of England, "Bank Rate (The Interest Rate)," Available at: <https://www.bankofengland.co.uk/monetary-policy/the-interest-rate-bank-rate>, Accessed: January 2025.
- [40] SolarGuide, "Tesla Powerwall 2.0 Cost, Specs and Reviews," <https://www.solarguide.co.uk/tesla-energy/powerwall-2>, accessed: January 2025.
- [41] F. C. Melo, G. Carrilho da Graça, and M. J. Oliveira Panão, "A review of annual, monthly, and hourly electricity use in buildings," *Energy and Buildings*, vol. 293, p. 113201, 2023.
- [42] J. Heinonen and S. Junnila, "Residential energy consumption patterns and the overall housing energy requirements of urban and rural households in finland," *Energy and Buildings*, vol. 76, pp. 295–303, 2014.
- [43] Ministry of Housing, Communities and Local Government, "Size of English Homes Fact Sheet: English Housing Survey 2018-19," 2020, Accessed: April 2025.
- [44] B. Zakeri, G. C. Gissey, P. E. Dodds, and D. Subkhankulova, "Centralized vs. distributed energy storage – benefits for residential users," *Energy*, vol. 236, p. 121443, 2021.
- [45] British Gas, "Average Gas and Electricity Bills Explained," 2025, Accessed: April 2025. [Online]. Available: <https://www.britishgas.co.uk/energy/guides/average-bill.html>